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Limits: Direct Substitution & LHL/RHL

50 Objective Questions

Select answer → check & get solution

Question 1

$$\lim_{x \to 2}(3x^2 - 5x + 1)$$

- 3
- 5
- 7
- 9

Check Answer

Correct answer: C. 7

Solution: Direct substitution: $3(2)^2 - 5(2) + 1 = 12 - 10 + 1 = 3$.

Question 2

$$\lim_{x \to -1} \frac{x^2 - 1}{x + 1}$$

- ◯ -2
- ◯ 0
- ◯ 2
- ◯ $Doesnotexist$

Check Answer

💡 **Correct answer:** A. -2

Solution: Factor: $\frac{(x-1)(x+1)}{x+1} = x - 1$ for $x \neq -1$. Limit = $-1 - 1 = -2$.

Question 3

$$\lim_{x \to 0} \frac{\sin 3x}{x}$$

- ◯ 0
- ◯ 1
- ◯ 3
- ◯ ∞

Check Answer

💡 **Correct answer:** C. 3

Solution: Standard limit: $\lim_{x \to 0} \frac{\sin kx}{x} = k$. Here $k = 3$.

Question 4

$$\lim_{x \to 0^+} \frac{|x|}{x}$$

- -1
- 0
- 1
- $Doesnotexist$

Check Answer

💡 **Correct answer:** C. 1

Solution: For $x > 0$, $|x| = x$, so $\frac{x}{x} = 1$. Right-hand limit = 1.

Question 5

$$\lim_{x \to 0^-} \frac{|x|}{x}$$

- -1
- 0
- 1
- $Doesnotexist$

Check Answer

💡 **Correct answer:** A. -1

Solution: For $x < 0$, $|x| = -x$, so $\frac{-x}{x} = -1$. Left-hand limit = -1.

Question 6

$$\lim_{x \to 0} \frac{|x|}{x}$$

- ◯ -1
- ◯ 0
- ◯ 1
- ◯ $Doesnotexist$

Check Answer

💡 **Correct answer:** D. $Doesnotexist$
Solution: LHL = -1, RHL = 1, not equal → limit does not exist.

Question 7

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

- ◯ 0
- ◯ 2
- ◯ 4
- ◯ 6

Check Answer

💡 **Correct answer:** C. 4
Solution: Factor: $\frac{(x-2)(x+2)}{x-2} = x + 2 \to 4$.

Question 8

$$\lim_{x \to 1} \frac{\sqrt{x} - 1}{x - 1}$$

- ◯ $\frac{1}{2}$
- ◯ 1
- ◯ 0
- ◯ ∞

Check Answer

💡 **Correct answer:** A. $\frac{1}{2}$

Solution: Rationalise: $\frac{1}{\sqrt{x}+1} \to \frac{1}{2}$.

Question 9

$$\lim_{x \to 0} \frac{e^x - 1}{x}$$

- ◯ 0
- ◯ 1
- ◯ e
- ◯ ∞

Check Answer

💡 **Correct answer:** B. 1

Solution: Standard limit: $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$.

Question 10

$$\lim_{x \to 0} \frac{\ln(1+x)}{x}$$

- 0
- 1
- e
- ∞

Check Answer

Correct answer: B. 1

Solution: Standard limit: $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$

Question 11

$$\lim_{x \to 0} \frac{\tan x}{x}$$

- 0
- 1
- ∞
- $Doesnotexist$

Check Answer

Correct answer: B. 1

Solution: $\lim_{x \to 0} \frac{\tan x}{x} = 1.$

Question 12

$$\lim_{x \to 3}(2x^3 - 4x + 5)$$

- ◯ 41
- ◯ 45
- ◯ 47
- ◯ 49

Check Answer

💡 **Correct answer:** C. 47
Solution: Direct substitution: $2(27) - 12 + 5 = 47$.

Question 13

$$\lim_{x \to 0} \frac{\sin^2 x}{x^2}$$

- ◯ 0
- ◯ 1
- ◯ 2
- ◯ ∞

Check Answer

💡 **Correct answer:** B. 1
Solution: $\lim_{x \to 0} \frac{\sin x}{x} = 1$, square gives 1.

Question 14

$$\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$$

- ◯ 6
- ◯ 8
- ◯ 12
- ◯ 0

Check Answer

💡 **Correct answer:** C. 12
Solution: $x^2 + 2x + 4 \to 12$.

Question 15

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2}$$

- ◯ 0
- ◯ $\frac{1}{2}$
- ◯ 1
- ◯ 2

Check Answer

💡 **Correct answer:** B. $\frac{1}{2}$
Solution: Standard limit = 1/2.

Question 16

$$\lim_{x \to 1} \frac{x^n - 1}{x - 1}$$

- ◯ n
- ◯ $n - 1$
- ◯ 1
- ◯ 0

Check Answer

💡 **Correct answer:** A. n
Solution: Derivative definition: n.

Question 17

$$\lim_{x \to 0} \frac{\sin^{-1} x}{x}$$

- ◯ 0
- ◯ 1
- ◯ ∞
- ◯ $\pi/2$

Check Answer

💡 **Correct answer:** B. 1
Solution: Standard limit = 1.

Question 18

$$\lim_{x \to 0} \frac{\tan^{-1} x}{x}$$

- 0
- 1
- ∞
- $\pi/4$

Check Answer

💡 **Correct answer:** B. 1
Solution: Standard limit = 1.

Question 19

$$\lim_{x \to 2} \frac{\sqrt{x+2} - 2}{x - 2}$$

- $\frac{1}{4}$
- $\frac{1}{2}$
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\frac{1}{4}$
Solution: Rationalise: $\frac{1}{\sqrt{x+2}+2} \to \frac{1}{4}$.

Question 20

$$\lim_{x \to 0} \frac{\sqrt{1+x}-1}{x}$$

- 0
- $\frac{1}{2}$
- 1
- ∞

Check Answer

💡 **Correct answer:** B. $\frac{1}{2}$

Solution: Rationalise → 1/2.

Question 21

$$\lim_{x \to 0} \frac{\sin 2x}{\sin 3x}$$

- $\frac{2}{3}$
- $\frac{3}{2}$
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\frac{2}{3}$

Solution: $\frac{2}{3}$.

Question 22

$$\lim_{x \to 0} \frac{e^{2x} - 1}{\sin 3x}$$

- ◯ $\frac{2}{3}$
- ◯ $\frac{3}{2}$
- ◯ 2
- ◯ 3

Check Answer

💡 **Correct answer:** A. $\frac{2}{3}$

Solution: $\frac{2}{3}$.

Question 23

$$\lim_{x \to 1} \frac{x^2 - 1}{x^3 - 1}$$

- ◯ $\frac{1}{2}$
- ◯ $\frac{2}{3}$
- ◯ $\frac{3}{2}$
- ◯ 2

Check Answer

💡 **Correct answer:** B. $\frac{2}{3}$

Solution: $\frac{2}{3}$.

Question 24

$$\lim_{x \to 0} \frac{\ln(1+2x)}{x}$$

- 0
- 1
- 2
- ∞

Check Answer

💡 **Correct answer:** C. 2
Solution: 2.

Question 25

$$\lim_{x \to 0} \frac{3^x - 1}{x}$$

- $\ln 3$
- 3
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\ln 3$
Solution: $\ln 3$.

Question 26

$$\lim_{x \to 0} \frac{1 - \cos 2x}{x^2}$$

- 1
- 2
- 4
- 0

Check Answer

💡 **Correct answer:** B. 2
Solution: 2.

Question 27

$$\lim_{x \to 0^+} \frac{1}{x}$$

- $-\infty$
- 0
- 1
- $+\infty$

Check Answer

💡 **Correct answer:** D. $+\infty$
Solution: $+\infty$.

Question 28

$$\lim_{x \to 0^-} \frac{1}{x}$$

- $-\infty$
- 0
- 1
- $+\infty$

Check Answer

💡 **Correct answer:** A. $-\infty$
Solution: $-\infty$.

Question 29

$$\lim_{x \to 2} \frac{|x-2|}{x-2}$$

- -1
- 0
- 1
- $Doesnotexist$

Check Answer

💡 **Correct answer:** D. $Doesnotexist$
Solution: LHL=-1, RHL=1 → DNE.

Question 30

$$\lim_{x \to 1} \frac{x^2 - 3x + 2}{x - 1}$$

- -1
- 0
- 1
- 2

Check Answer

💡 **Correct answer:** A. -1
Solution: $x - 2 \to -1$.

Question 31

$$\lim_{x \to 0} \frac{\tan x - x}{x^3}$$

- $\frac{1}{3}$
- $\frac{1}{2}$
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\frac{1}{3}$
Solution: Series → 1/3.

Question 32

$$\lim_{x \to 0} \frac{x - \sin x}{x^3}$$

- $\frac{1}{6}$
- $\frac{1}{3}$
- $\frac{1}{2}$
- 0

Check Answer

💡 **Correct answer:** A. $\frac{1}{6}$

Solution: Series → 1/6.

Question 33

$$\lim_{x \to 0} \frac{\sin x - x}{x^3}$$

- $-\frac{1}{6}$
- $\frac{1}{6}$
- $-\frac{1}{3}$
- $\frac{1}{3}$

Check Answer

💡 **Correct answer:** A. $-\frac{1}{6}$

Solution: Series → -1/6.

Question 34

$$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$$

- $\frac{1}{2}$
- 0
- 1
- ∞

Check Answer

💡 **Correct answer:** A. $\frac{1}{2}$

Solution: Series → 1/2.

Question 35

$$\lim_{x \to 0} \frac{\ln(1+x) - x}{x^2}$$

- ◯ $-\frac{1}{2}$
- ◯ $\frac{1}{2}$
- ◯ 0
- ◯ 1

Check Answer

💡 **Correct answer:** A. $-\frac{1}{2}$
Solution: Series → -1/2.

Question 36

$$\lim_{x \to 0} (1+x)^{1/x}$$

- ◯ e
- ◯ 1
- ◯ ∞
- ◯ 0

Check Answer

💡 **Correct answer:** A. e
Solution: Standard limit = e.

Question 37

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$$

- e
- 1
- ∞
- 0

Check Answer

💡 **Correct answer:** A. e
Solution: Standard limit = e.

Question 38

$$\lim_{x\to 0}\frac{\sqrt{1+x^2}-1}{x^2}$$

- $\frac{1}{2}$
- 1
- 0
- ∞

Check Answer

💡 **Correct answer:** A. $\frac{1}{2}$
Solution: Rationalise → 1/2.

Question 39

$$\lim_{x \to 0} \frac{\tan^{-1} x - x}{x^3}$$

- $-\frac{1}{3}$
- $\frac{1}{3}$
- $-\frac{1}{2}$
- $\frac{1}{2}$

Check Answer

💡 **Correct answer:** A. $-\frac{1}{3}$

Solution: Series → -1/3.

Question 40

$$\lim_{x \to 0} \frac{\sin 5x}{\tan 3x}$$

- $\frac{5}{3}$
- $\frac{3}{5}$
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\frac{5}{3}$

Solution: $5/3$.

Question 41

$$\lim_{x \to 2} \frac{x^2 - 4}{x^3 - 8}$$

- ○ $\frac{1}{3}$
- ○ $\frac{1}{2}$
- ○ $\frac{2}{3}$
- ○ 1

Check Answer

💡 **Correct answer:** A. $\frac{1}{3}$

Solution: $\frac{x+2}{x^2+2x+4} \to \frac{4}{12} = 1/3.$

Question 42

$$\lim_{x \to 0} \frac{\sin^{-1} x - x}{x^3}$$

- ○ $\frac{1}{6}$
- ○ $-\frac{1}{6}$
- ○ $\frac{1}{3}$
- ○ $-\frac{1}{3}$

Check Answer

💡 **Correct answer:** A. $\frac{1}{6}$

Solution: Series → 1/6.

Question 43

$$\lim_{x \to 0} \frac{1 - \cos x}{x \sin x}$$

- ○ $\frac{1}{2}$
- ○ 1
- ○ 0
- ○ ∞

Check Answer

💡 **Correct answer:** A. $\frac{1}{2}$

Solution: $\frac{1}{2}$.

Question 44

$$\lim_{x \to 0} \frac{\sin x - \tan x}{x^3}$$

- $-\frac{1}{2}$
- $\frac{1}{2}$
- $-\frac{1}{3}$
- $\frac{1}{3}$

Check Answer

💡 **Correct answer:** A. $-\frac{1}{2}$

Solution: Series → -1/2.

Question 45

$$\lim_{x \to 0} \frac{x - \tan x}{x^3}$$

- ◯ $-\frac{1}{3}$
- ◯ $\frac{1}{3}$
- ◯ $-\frac{1}{2}$
- ◯ $\frac{1}{2}$

Check Answer

💡 **Correct answer:** A. $-\frac{1}{3}$
Solution: Series → -1/3.

Question 46

$$\lim_{x \to 0} \frac{e^x - e^{-x}}{x}$$

- ◯ 0
- ◯ 1
- ◯ 2
- ◯ ∞

Check Answer

💡 **Correct answer:** C. 2
Solution: 2.

Question 47

$$\lim_{x \to 0} \frac{\cos x - 1}{x^2}$$

- $-\frac{1}{2}$
- $\frac{1}{2}$
- 0
- 1

Check Answer

💡 **Correct answer:** A. $-\frac{1}{2}$

Solution: $-1/2$.

Question 48

$$\lim_{x \to 0} \frac{\sin x - x}{x^3}$$

- $-\frac{1}{6}$
- $\frac{1}{6}$
- $-\frac{1}{3}$
- $\frac{1}{3}$

Check Answer

💡 **Correct answer:** A. $-\frac{1}{6}$

Solution: $-1/6$.

Question 49

$$\lim_{x \to 0} \frac{\tan x - \sin x}{x^3}$$

- $\frac{1}{2}$
- $\frac{1}{3}$
- 1
- 0

Check Answer

💡 **Correct answer:** A. $\frac{1}{2}$

Solution: $1/2$.

Tap "Check Answer" for individual feedback, or "See All Solutions" to reveal all answers at once.